Marilyn Rego

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RESEARCH INTERESTS

• Distributed AI Systems • Communication-Efficient Training & Inference • Heterogeneous Computing • Performance Engineering • Performance Analysis & Optimization • Hardware–Software Co-Design

EDUCATION

University of Michigan — Ann Arbor, MI
Ph.D. in Computer Science & Engineering — Aug 2025 - May 2030
Advisor: Muhammad Shahbaz

Purdue University — West Lafayette, IN
Bachelor of Science, Computer Science GPA 3.9/4.0 — Aug 2021 - May 2025
Concentration in Machine Intelligence | Minors: Mathematics, Accounting, Management

PUBLICATIONS

Conference Papers

ISSRE. Wen Fan, Minh Tran, Sanya Dod, Xin Hu, **Marilyn Rego**, Danning Xie, Jenna DiVincenzo, and Lin Tan. *An Empirical Study of LLM-Generated Specifications for VeriFast.* September 2026.

FORGE. Marilyn Rego*, Wen Fan*, Xin Hu, Sanya Dod, Zhaorui Ni, Danning Xie, Jenna DiVincenzo, and Lin Tan. *Evaluating the Ability of Large Language Models to Generate Verifiable Specifications in VeriFast.* November 2025. (*co-primary author)

Workshop Papers

NAIC Workshop @ SIGCOMM. Regan McDonald, **Marilyn Rego,** Ertza Warraich, Annus Zulfiqar, and Muhammad Shahbaz. *Towards Network-Efficient Cross-Regional Inference via Learned Activation Compression.* August 2026.

SELECTED PRESENTATIONS

NSDI. Marilyn Rego, Maxwell Kumbong, Hermann Kumbong, Ertza Warraich, and Muhammad Shahbaz. mrLLM: Fast Multi-Region LLM Inference using Learned Adapters. May 2026. (Poster)

SRC TECHCON. Marilyn Rego. SpliDT: Partitioned Decision Trees for Scalable Stateful ML Inference at Line Rate. September 2025. (Talk & Poster)

Purdue Summer Undergraduate Research Symposium. Marilyn Rego. AI for Software Verification. July 2024. (Talk)

AWARDS AND HONOR

- Adobe Research Code Quality Jam – Robustness Category Winner — 2026
- NSDI'26 Student Grant Award — 2026
- First-Year CSE PhD Rackham Fellowship — 2025
- NSDI'25 Student Grant Award — 2025
- Purdue College of Science Dean's List and Semester Honors (All Semesters) — 2021-2025

RESEARCH & PROFESSIONAL EXPERIENCE

Adobe — Seattle, WA
Research Scientist/Engineer Intern — May 2026 – August 2026

- Developed a heterogeneous CPU–GPU performance profiling framework for distributed AI systems, enabling unified analysis across CPUs, GPUs, and multi-process execution.
- Designed cross-device profiling and synchronization techniques to improve performance analysis for GPU-

accelerated applications.
- Enabled end-to-end profiling of Python, PyTorch, multi-GPU, and vLLM-based LLM serving workloads.
- Enhanced the reliability and scalability of the profiling infrastructure through validation and platform compatibility improvements.

SRC Research Scholars Program Ann Arbor, MI
Graduate Student Research Assistant Feb 2025 - Present
- Conducting research on communication-efficient systems for wide-area distributed LLM training and inference through hardware–software co-design, targeting scalable execution across geographically distributed data centers.

Purdue University, Department of Computer Science, NextGArchLab West Lafayette, IN
Undergraduate Student Research Assistant Jan 2024 – August 2025
Advisor: Muhammad Shahbaz
- Developing and optimizing ensemble machine learning models for network classification in PISA switches, with a focus on improving performance and accuracy.
- Implemented and benchmarked classification models (One-class SVMs, Random Forest, Isolation Forest) in Python to analyze packet payload features for malicious vs benign traffic, establishing baseline accuracy, computational cost, and scalability trade-offs for real-world security applications.
- Designing algorithms that enable efficient, data-driven decision-making in modern computer networks.

Purdue University, School of Electrical and Computer Engineering West Lafayette, IN
Summer Undergraduate Research Fellowship May 2024 – August 2024
Advisor: Dr. Jenna DiVincenzo
- Evaluated open-source large language models (LLMs) for their ability to generate fully accurate specifications for static verification, applying techniques like prompt engineering, chain-of-thought (CoT), and few-shot prompting to assess model effectiveness.
- Categorized VeriFast benchmarks and developed tailored input datasets for training and testing LLMs.
- Designed qualitative analysis methodologies to quantify the quality of LLM outputs and conducted qualitative result interpretation for research and workshop publications.

Purdue University, Department of Chemistry West Lafayette, IN
Undergraduate Research Assistant May 2023 – August 2023
Advisor: Dr. Tonglei Li
- Implemented a Python program for predicting 3-dimensional molecular structure and their properties based on user-input parameters in the project – "Deep Learning for Drug Discovery".
- Optimized the program using ML algorithms to decipher molecular mechanisms of crystal growth and improve prediction accuracy.

PROFESSIONAL SERVICE

- Judge, University of Michigan Undergraduate Research Symposium 2026

MENTORING EXPERIENCE

- Maxwell Kumbong, SWE @ Duolingo – Distributed AI Systems 2025 - Present
- Yuchen Wang, undergrad researcher at University of Michigan – Distributed AI Systems 2026
- Regan McDonald, undergrad researcher at University of Michigan – LLM Activation Compression 2025 - 2026

TEACHING EXPERIENCE

Purdue University. CS 182 – Foundations of Computer Science (UTA). Spring 2023